Bootstrapping the Yield Curve: Coupon-Bond Yields vs. Spot Rates

FINC409/609: Fixed Income Securities
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This handout gives a numerical example showing why fixed-income valuation should be based on *bootstrapped discount factors* (or, equivalently, spot rates), rather than on the yields to maturity of coupon bonds. Throughout, assume: Face value is 100, coupons are paid annually, compounding is annual, and there is no default risk.

1. The basic pricing principle

If a bond has cash flows $C_1, C_2, \ldots, C_n$ at dates $1, 2, \ldots, n$, then the its price is

$$P = \sum_{t=1}^{n} C_t \, D(t).$$

Here $D(T)$ denotes the discount factor for maturity T. If $r(t)$ denotes the t-year spot rate, then

$$D(t) = \frac{1}{(1 + r(t))^t}, \qquad r(t) = D(t)^{-1/t} - 1.$$

The key point is that each cash flow must be discounted using the discount factor for its own maturity. A coupon-bond yield to maturity is *not* a discount rate for an individual cash flow. It is a single internal rate of return that prices the bond's *entire* cash-flow stream.

2. Market quotes

Suppose the following coupon bonds are observed in the market, and each trades at par:

Maturity	Coupon rate	Price	Yield to maturity
1 year	2%	100	2%
2 years	5%	100	5%
3 years	8%	100	8%
4 years	11%	100	11%
5 years	14%	100	14%

Because each bond trades at par, its yield to maturity equals its coupon rate. These observed yields are therefore **par yields**. They are not spot rates.

3. Bootstrapping the discount factors

Because face value is 100, the coupon payment equals the coupon rate in dollars. For example, the 3-year bond with an 8% coupon pays 8 each year and 108 at maturity.

Step 1: Solve for $D(1)$

The 1-year bond pays 102 at time 1 and has price 100, so

$$100 = 102\,D(1).$$

Hence

$$D(1) = \frac{100}{102} = 0.980392.$$

Step 2: Solve for $D(2)$

The 2-year bond pays 5 at time 1 and 105 at time 2, so

$$100 = 5\,D(1) + 105\,D(2).$$

Therefore

$$D(2) = \frac{100 - 5D(1)}{105} = \frac{100 - 5(0.980392)}{105} = 0.905696.$$

Step 3: Solve for $D(3)$

The 3-year bond pays 8 at times 1 and 2, and 108 at time 3:

$$100 = 8\,D(1) + 8\,D(2) + 108\,D(3).$$

So

$$D(3) = \frac{100 - 8D(1) - 8D(2)}{108} = 0.786216.$$

Step 4: Solve for $D(4)$

The 4-year bond pays 11 at times 1, 2, and 3, and 111 at time 4:

$$100 = 11\,D(1) + 11\,D(2) + 11\,D(3) + 111\,D(4).$$

Thus

$$D(4) = \frac{100 - 11D(1) - 11D(2) - 11D(3)}{111} = 0.636078.$$

Step 5: Solve for $D(5)$

The 5-year bond pays 14 at times 1, 2, 3, and 4, and 114 at time 5:

$$100 = 14\,D(1) + 14\,D(2) + 14\,D(3) + 14\,D(4) + 114\,D(5).$$

Hence

$$D(5) = \frac{100 - 14D(1) - 14D(2) - 14D(3) - 14D(4)}{114} = 0.470901.$$

Bootstrapped discount factors and spot rates

Using

$$r(t) = D(t)^{-1/t} - 1,$$

we obtain the following spot rates:

Maturity t	Discount factor $D(t)$	Spot rate $r(t)$
1	0.980392	2.000%
2	0.905696	5.077%
3	0.786216	8.348%
4	0.636078	11.975%
5	0.470901	16.256%

This table already shows the source of the problem: the 5-year coupon bond yield is 14%, but the bootstrapped 5-year spot rate is about 16.256%. The coupon-bond yield is not the correct discount rate for a 5-year cash flow.

4. Correctly pricing a new bond

Now consider a new 5-year bond with a 10% annual coupon. Its cash flows are

$$10,\ 10,\ 10,\ 10,\ 110.$$

Using the bootstrapped discount factors, the correct price is

$$\begin{aligned} P_{\text{correct}} &= 10D(1) + 10D(2) + 10D(3) + 10D(4) + 110D(5) \\ &= 10(0.980392) + 10(0.905696) + 10(0.786216) + 10(0.636078) + 110(0.470901) \\ &= 84.883. \end{aligned}$$

This is the no-arbitrage value implied by the bootstrapped term structure.

5. The incorrect shortcut

A common mistake is to take the observed coupon-bond yields, namely

$$2\%,\ 5\%,\ 8\%,\ 11\%,\ 14\%,$$

and treat them as though they were the 1-year, 2-year, 3-year, 4-year, and 5-year spot rates.

Under this incorrect approach, the bond would be valued as

$$\begin{aligned} P_{\text{wrong}} &= \frac{10}{1.02} + \frac{10}{(1.05)^2} + \frac{10}{(1.08)^3} + \frac{10}{(1.11)^4} + \frac{110}{(1.14)^5} \\ &= 90.530. \end{aligned}$$

Therefore, the pricing error is

$$P_{\text{wrong}} - P_{\text{correct}} = 90.530 - 84.883 = 5.648.$$

This is a large error for a bond with face value 100. In percentage terms, the bond is overpriced by about 6.65% relative to the correct value.

6. Why the mistake is so large

The error is large because the yield to maturity on a coupon bond is a weighted average of discount rates relevant for that bond's entire cash-flow stream. It is not a maturity-specific spot rate.

In this example, the final cash flow of the 10% bond is 110 at time 5, so a large fraction of the bond's present value depends on the discount factor at the 5-year horizon. The incorrect shortcut discounts this payment using 14%, while the correct bootstrapped 5-year spot rate is 16.256%.

Equivalently,

$$\frac{1}{(1.14)^5} = 0.519369, \qquad D(5) = 0.470901.$$

The incorrect method therefore assigns too much present value to the largest cash flow. That is why it materially overprices the bond.

7. Key Takeaway

Coupon-bond yields and par yields should not be used as discount rates for individual cash flows. To value a bond correctly, one must first bootstrap discount factors (or spot rates) and then discount each cash flow using the term structure appropriate to its own maturity.

A further implication is that two different 5-year coupon bonds can have different yields to maturity under the same spot curve because the timing weights on cash flows differ across bonds. Hence there is no single "5-year coupon-bond yield" that can correctly price every 5-year bond unless the term structure is flat.

All numerical values are rounded to three decimals for prices and to six decimals for discount factors.